DO FULL-DUPLEX SPEECH DIALOGUE MODELS KNOW WHEN TO ASK? MEASURING AND INSTILLING CLARIFICATION UNDER ACOUSTIC UNCERTAINTY

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ABSTRACT

Cascaded spoken dialogue systems have long used ASR confidence to decide when to confirm or ask the user to repeat. End-to-end full-duplex models such as Moshi expose no such confidence signal, and it is unknown whether they can convert acoustic uncertainty into the communicative act of asking. We present the first benchmark and closed-loop protocol for acoustically-grounded clarification: slot spans of assistant-directed utterances (MASSIVE, and real recordings from SLURP) are locally corrupted, and a frame-synchronous driver injects a scripted repair turn only when the model asks. Across public full-duplex models (Moshi variants and PersonaPlex), we find that models rarely ask and instead proceed with hallucinated slot values. We then show that LoRA fine-tuning on synthetic dialogues containing acoustic minimal pairs—identical text, clean→confirm versus corrupted→ask—instils clarification that is calibrated to degradation level, transfers from synthetic to real speech, and does not regress turn-taking, whereas a lexical-only control does not generalize to acoustic corruption.

Index Terms— spoken dialogue systems, full-duplex, clarification requests, uncertainty, speech language models

1. INTRODUCTION

Full-duplex spoken dialogue models have learned when to speak: end-to-end systems such as Moshi [1] model both conversation channels jointly and handle overlap, backchannels, and interruptions with human-like latency, and a rapidly growing family of open-weight successors—PersonaPlex [2], J-Moshi [3], and others—has made this behaviour widely reproducible. Benchmarks and challenges have followed the same trajectory, evaluating pause handling, backchanneling, and interruption timing [4, 5, 6]. What none of these systems or benchmarks address is whether a full-duplex model knows when it has failed to hear: whether, upon receiving an utterance whose task-critical content is acoustically unintelligible, the model can notice its own uncertainty and respond with the ordinary human remedy—asking.

Classical cascaded systems had this ability by construction. ASR confidence scores drove explicit confirmation and repair strategies, a standard component of deployed dialogue managers for two decades [7, 8], and recent work continues to refine cascade-side clarification by diagnosing error causes in ASR latent representations [9]. End-to-end full-duplex models dissolved the ASR interface, and with it the confidence signal that this machinery consumed. The implicit assumption is that the model will internalize the equivalent behaviour; whether it actually does has not been tested. The stakes are asymmetric: a system that silently substitutes a plausible but wrong slot value (“7 pm, got it”) is worse than one that fails, because the error is confident, fluent, and invisible to the user.

We answer this question with a measurement-then-intervention design. First, we construct a benchmark in which only the slot span of an assistant-directed utterance (e.g., the time in “wake me up at 5 pm tomorrow”) is corrupted—babble noise at graded SNR, masking, or low-pass filtering—leaving the carrier utterance intact, over both synthetic-speech carriers (MASSIVE [10]) and real close-talk recordings (SLURP [11]). A weak-ASR oracle verifies that each corruption actually destroys the span’s content. Second, because clarification only matters if the conversation can continue, we evaluate in a closed loop: a frame-synchronous driver monitors the model’s own text stream, detects turn ends, and injects a pre-synthesized repair utterance only when the model asks—entirely offline and reproducible, with no commercial API in the loop. Third, we intervene: we fine-tune with LoRA on synthetic dialogues whose central ingredient is acoustic minimal pairs—the same utterance appearing once clean (answered with a confirmation) and once span-corrupted (answered with a clarification request)—so that the only signal separating the two target behaviours is acoustic. A lexical-only control (clarify_lexical), trained to ask only on semantically underspecified inputs, isolates whether learned clarification is conditioned on acoustic evidence or merely imitates lexical cues, and an ASR-confidence cascade baseline locates the explicit upper bound that end-to-end models must approach.

Our contributions are: (1) a task definition and benchmark for acoustically-grounded clarification in full-duplex dialogue, with span-local corruption and oracle-verified recoverability over synthetic and real speech; (2) a fully automatic closed-loop evaluation protocol whose repair injection responds to model behaviour, with decision-quality metrics (clarification hit/false-alarm, hallucinated-confirmation rate, slot success after at most one repair); (3) a matched comparison of end-to-end models against a confidence-thresholded cascade on the clarification/false-alarm/task-success frontier, quantifying how far current open full-duplex models sit from an explicit-confidence reference; and (4) a fine-tuning recipe with acoustic minimal pairs that instils calibrated, acoustically-conditioned clarification without degrading turn-taking, with controls that support the causal claim (lexical-only training, mild-noise and position-matched artifact-negative confirmations, non-slot and global corruption probes, held-out corruption families, multi-seed replication).

2. RELATED WORK

Full-duplex spoken dialogue models. Moshi [1] introduced dual-stream speech-text modelling over the Mimi codec; its inner text stream doubles as a self-transcription, which our protocol exploits to avoid output ASR. Successors add persona and voice control [2], other languages [3], codec-free variants [12], RL-optimized interactivity [13], instruction-controlled conversational behaviour [14], and latent reasoning while listening [15]. All optimize when and how to speak; none addresses uncertainty-driven clarification.

Full-duplex evaluation. Full-Duplex-Bench v1/v1.5 [4] scores pause handling, backchannels, smooth turn-taking, and interruptions on fixed inputs; v2 [5] adds a multi-turn automated examiner (commercial API in the loop) for user-initiated corrections. The ICASSP 2026 HumDial challenge [6] consolidated turn-taking evaluation. Our protocol differs in both axis (model-initiated clarification under acoustic uncertainty) and mechanism (local, reproducible closed loop that reacts to the model's behaviour).

Clarification and uncertainty. Confidence-based confirmation is classical in cascaded SDS [7, 8]; recent cascade work diagnoses ASR-side error causes to choose clarification strategies [9]. On the end-to-end side, audio-aware LLMs' uncertainty has been probed in QA settings [16], and noise robustness is studied as silent tolerance—fusion against interference [17], cascades outperforming E2E under noise [18]—rather than as grounds for asking. We connect these threads: uncertainty must surface as a communicative act.

Resources. MASSIVE [10] (CC-BY, 51 locales) and SLURP [11] (real English recordings) share an intent/slot inventory, so identical metrics apply across synthetic and real carriers, and across languages for the generality study.

3. BENCHMARK AND CLOSED-LOOP PROTOCOL

3.1. Task

Each trial presents one assistant-directed utterance with an annotated target slot span. Under clean audio the correct behaviour is to proceed (confirm the slot value); under span corruption it is to issue a clarification request; after a scripted repair answer, to confirm the recovered value. Failure modes are hallucinated confirmation (proceeding with a wrong value), over-asking on clean input, and non-cooperative responses.

3.2. Span-local corruption

Slot spans are located by forced alignment (MMS-FA [19]) on both TTS and real audio; alignment fallbacks are recorded in the manifest and excluded from primary analysis. Corruptions are applied only inside the span: babble at SNR $\{+5, 0, -5, -10\}$ dB, masking by silence or noise, and lowpass at 800 Hz. Two placement controls disentangle what a decision responds to: a global condition (full_snr0) distributes the same noise energy over the whole utterance (overall quality vs. local loss), and a non-slot condition (nonslot_snr0) places the identical artifact on a duration-matched span outside the slot, where the correct behaviour is still to confirm—separating "an artifact is present, so ask" from "the task-critical content is gone, so ask" within the same utterance. All corruptions are deterministic functions of a per-case seed, so the benchmark is exactly reconstructable from code. A weak-ASR oracle (faster-whisper small, greedy) provides a manipulation check—the corrupted span must not be recoverable—and an independent difficulty axis for calibration analysis.

3.3. Closed-loop driver

The driver streams the stereo input to the model frame-synchronously and monitors the model's inner text stream, declaring a turn end after 1.2 s without text. A rule-based classifier over the model's own text decides online whether the turn is a clarification request; only then is the pre-synthesized repair utterance injected into the user channel, frame-aligned to the model's turn end. The online decision gates injection only; all reported classifications are recomputed offline, and agreement with an LLM judge is reported (κ) as a measurement-validity check. Because the model's text stream serves as its own transcript, no output ASR enters the loop, and the entire protocol runs offline on local GPUs.

3.4. Metrics

We report clarification-request rate (CRR) and targeted CRR (T-CRR: the request identifies the corrupted slot); hallucinated-confirmation rate (HCR); decision quality as hit rate (asking under corruption) versus false alarms (asking on clean), with balanced accuracy; calibration as CRR-vs-SNR monotonicity against oracle recoverability; slot success rate (SSR) after at most one repair exchange; and selective risk over the non-asking subset. Uncertainty is quantified with Wilson intervals and paired bootstrap over matched base utterances.

4. INSTILLING CALIBRATED CLARIFICATION

Our proposed recipe converts the measurement protocol into supervision. From the MASSIVE training split (disjoint surface forms from all evaluation sets; SLURP held out entirely) we generate synthetic two-channel dialogues with deterministic templates and TTS, then corrupt the user channel after synthesis, span-locally, exactly as in the benchmark.

Acoustic minimal pairs. The training set's central ingredient: each selected utterance appears twice with identical text and identical TTS rendering—once clean, answered by a confirmation, and once span-corrupted, answered by a targeted clarification request followed by the repair exchange. Because the text streams of the pair are identical, a model can only separate the two target behaviours by attending to the acoustics of the user channel. Removing the clean twin (ablation) tests whether the pairing, not mere exposure to corrupted audio, drives acoustic conditioning.

Shortcut controls. A quarter of confirmation-labelled examples carry mild corruption (babble at +5 dB), penalizing the degenerate policy "noise present   ask" and forcing the decision boundary to track recoverability rather than noise presence. A further artifact-negative subset places the full-strength artifact on a duration-matched non-slot span, still labelled confirm: a position-matched control that the mild-noise mixture alone cannot provide. Both are ablated; the predicted failure signature is a rise in clean- and non-slot-condition false alarms.

Variants. Three training sets differ only in supervision: task_only (all confirmations, no clarification) controls for task formatting; clarify_lexical asks only on semantically underspecified inputs (clean audio), giving the model every lexical cue for asking but no acoustic one; clarify_full is the proposed set. If clarification were lexical imitation, clarify_lexical would generalize to acoustic corruption; the acoustic-conditioning hypothesis predicts it does not. All variants are trained identically with LoRA on the same base model, and evaluated on synthetic (MASSIVE-TTS), real (SLURP), and cross-lingual benchmarks, with turn-taking regression checked on Full-Duplex-Bench-derived metrics.

An ASR-confidence cascade (faster-whisper with a swept decision threshold) provides the explicit-confidence reference: its ROC locates what a system with a confidence signal achieves, and each end-to-end model appears as an operating point against it.

5. EXPERIMENTS

Experimental results are being produced on the PBS grid described in the project design document; this section is intentionally left as a stub in the current draft.

6. CONCLUSION

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